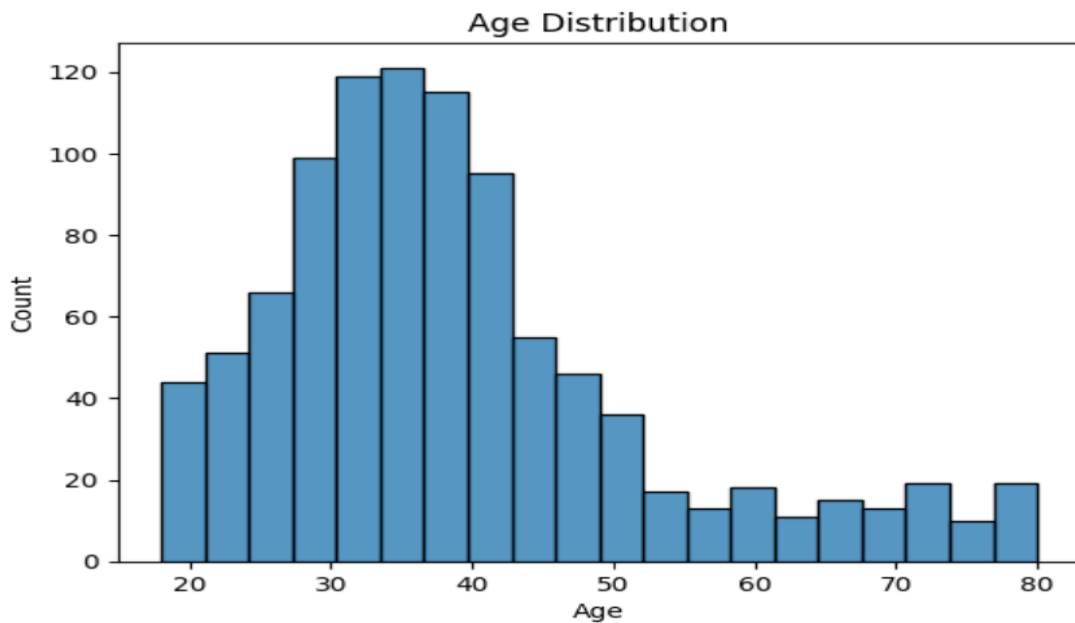
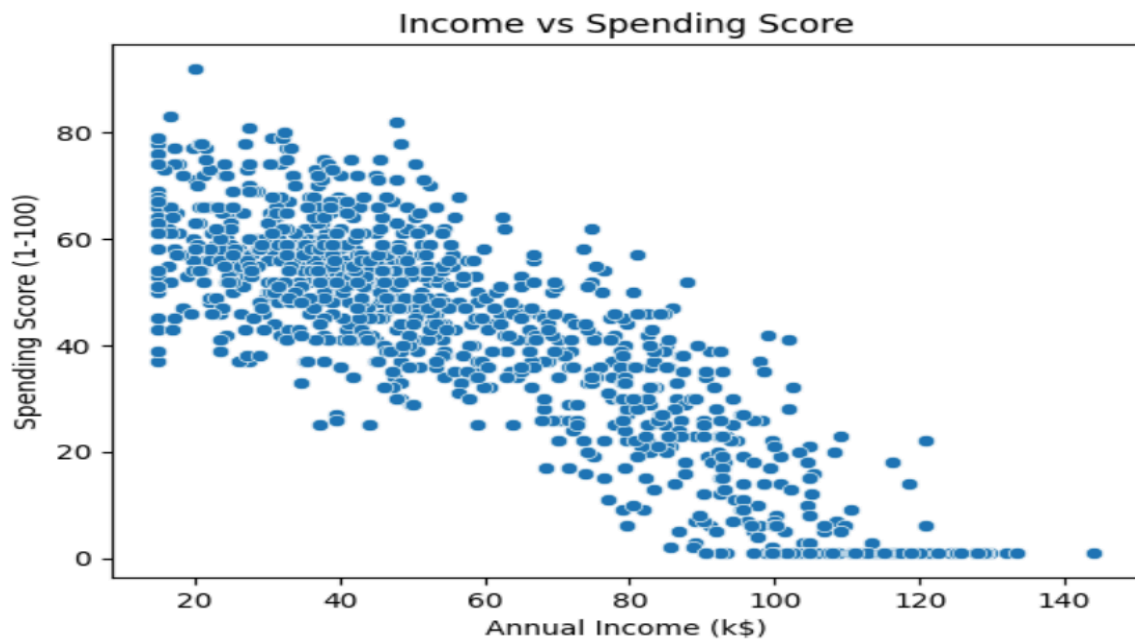


```
sns.histplot(df['Age'], bins=20)
plt.title("Age Distribution")
plt.show()
```



```
sns.scatterplot(x='Annual Income (k$)', y='Spending Score (1-100)',
data=df)
plt.title("Income vs Spending Score")
plt.show()
```

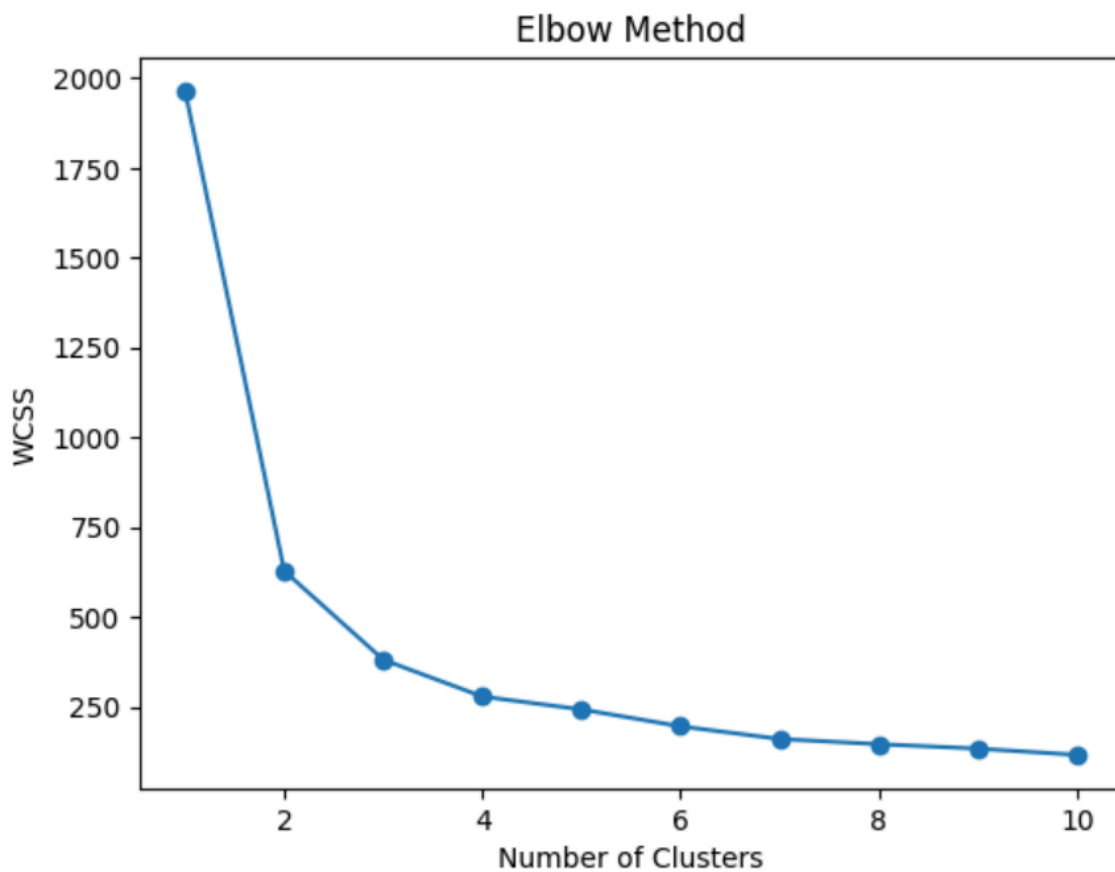


```
from sklearn.cluster import KMeans

wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```



```
plt.figure(figsize=(8,6))

plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'],
            c=df['Cluster'], cmap='viridis')

plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Customer Segments")

plt.show()
```

